

Evaluation and Validation Plan

The objective of this project is to identify socioeconomic vulnerability patterns across Toronto neighbourhoods and develop a transparent analytical framework that supports municipal decision-making. To ensure methodological rigor, multiple validation strategies will be applied throughout the analysis.

Principal Component Analysis (PCA) Validation

Principal Component Analysis (PCA) will be used to reduce dimensionality and address multicollinearity among socioeconomic indicators. PCA effectiveness will be evaluated using:

- Scree Plot analysis to identify the optimal number of components.
- Cumulative Explained Variance Ratio.
- Component loading interpretation to ensure retained components remain meaningful and interpretable.
- Retention of components explaining approximately 80–90% of total variance.

The selected components will be compared against the original variables to confirm that important socioeconomic information is preserved.

K-Means Clustering Validation

K-Means clustering will be used to identify neighbourhood groups with similar socioeconomic characteristics.

Cluster quality will be evaluated using:

- Elbow Method to determine the optimal number of clusters.
- Silhouette Score to measure cluster separation and cohesion.
- Cluster stability analysis by comparing clustering results across different values of k .
- Visual inspection using PCA-based cluster visualizations.

Clusters will be interpreted based on their socioeconomic characteristics to ensure practical relevance for policy planning.

Predictive Model Validation

Future project phases will incorporate historical neighbourhood profile data from earlier census periods (for example 2006 & 2011) to support predictive modelling.

Predictive models will be evaluated using:

- Accuracy
- Precision

- Recall
- F1 Score
- ROC-AUC

A training-validation-testing will be implemented using either:

- 80/20 Train-Test Split, or
- 5-Fold Cross Validation

The final approach will depend on dataset size and class distribution.

Longitudinal Analysis Validation

For temporal analysis using multiple census years, validation will focus on:

- Consistency of variable definitions across census years.
- Trend comparisons between neighbourhoods.
- Stability of identified vulnerability patterns over time.
- Sensitivity analysis to evaluate the impact of missing data and changing neighbourhood characteristics.

Assumptions, Leakage Prevention, and Fair Comparison Controls

PCA assumes:

- Variables are numeric and continuous.
- Relationships between variables are approximately linear.
- Variables are standardized before analysis.
- Sufficient correlation exists among variables.
- Components capture meaningful variation within the data.

To satisfy these assumptions, all socioeconomic indicators will be standardized using z-score normalization prior to PCA.

K-Means assumes:

- Clusters are relatively spherical in shape.
- Variables are continuous.
- Euclidean distance appropriately represents similarity.
- Features are measured on comparable scales.

Standardization will be applied before clustering to prevent variables with larger scales from dominating cluster formation.

Leakage Prevention Strategy

Data leakage occurs when information unavailable during prediction unintentionally influences model development.

Target Leakage- Variables directly derived from Low Income Prevalence may artificially inflate predictive performance.

Mitigation:

- Carefully review all candidate variables.
- Exclude variables representing the same underlying concept as the target.

Preprocessing Leakage- Applying scaling, PCA, or feature selection before data splitting can expose testing information to the model.

Mitigation:

- Split data before preprocessing.
- Fit scaling and PCA only on training data.
- Apply transformations to validation and test data using training parameters.

Temporal Leakage - Future neighbourhood information must not be used when predicting earlier periods.

Mitigation:

- Historical data will be used chronologically.
- Training datasets will contain only information available at the prediction date.

Duplicate Record Leakage- Duplicate neighbourhood records may bias results.

Mitigation:

- Perform duplicate detection during data cleaning.
- Remove duplicate observations prior to analysis.

Fair Comparison Controls

When comparing alternative approaches, all methods will be evaluated using:

- Identical datasets.
- Identical preprocessing procedures.
- Identical evaluation metrics.
- Identical train-test splits.

Changes Implemented Following Milestone 1 Feedback

The current project reflects several revisions made in response to faculty professor feedback received during Milestone 1.

Milestone 1 Feedback	Action Implemented
Clarify analyst role and decision context	Defined the project as a municipal socioeconomic analytics study supporting city planners, housing analysts, and community organizations.
Narrow and strengthen research questions	Refined the project into 3 focused research questions addressing vulnerability drivers, neighbourhood classification, and future vulnerability trends.
Improve dataset documentation	Expanded dataset biography to include source, purpose, variables, structure, coverage, and limitations.
Strengthen ethics discussion	Added privacy, bias, representativeness, and responsible-use considerations.
Conduct exploratory data analysis	Completed data profiling, variable assessment, correlation analysis, and dimensionality reduction preparation.
Improve technical justification	Added methodological rationale explaining why PCA and clustering were selected over simpler alternatives.
Develop reproducible workflow	Established GitHub repository with organized folders for data, notebooks, reports, and visualizations.
Plan for predictive modelling	Identified future integration of historical Toronto neighbourhood datasets from earlier census periods to support longitudinal analysis and predictive modelling.

These revisions strengthen project feasibility, methodological transparency, reproducibility, and alignment with stakeholder decision-making requirements. The project has evolved from an initial proposal into a structured analytical framework suitable for advanced socioeconomic vulnerability analysis.